No-load acceleration from 90% to 100% speed
Five-second target ramp: frozen-controller comparison

T-MATS simulation study

10 October 2026

Abstract

Four existing controllers are tested on the same nonlinear plant with zero external shaft extraction. The target rises from 9000 to 10000 rpm in five seconds. Base PI has the lowest ramp tracking error in this experiment. Ringing-aware PILCO improves ramp tracking relative to the previous PILCO, but does not outperform base PI. This is a transfer test of frozen policies, not a new training result.

1 Identical-sensor audit and fresh rerun

All four models explicitly use the same T-MATS first-order speed-sensor library block: time constant 0.05 s, unity gain, limits 0–20000 rpm, initial value from the same plant initial speed, noise disabled, unity fault gain and zero drift. The sensor sample-time setting is the same plant solve interval. Model wiring assertions verify that both the base PI and each active advanced controller receive this sensor output directly. The sensor input comes from the same plant-output block in every model.

The earlier models already had these settings. The new simulations explicitly enforce them in separate model copies and reproduce the earlier speed, sensed-speed and fuel traces. Thus the prior ranking was not caused by unequal sensor dynamics. Controllers respond differently to the same lag. During a ramp, actual-speed error and sensed-speed error differ; the primary tables continue to score actual shaft speed. Individual audit JSON files and a sensor-comparison CSV accompany the results.

2 Reproducible test protocol

Rated speed is defined as $N_{100} = 10000$ rpm. With $\tau = t - 60$ s, the reference is

$$r(\tau) = 9000 + 200 \min\{5, \max\{0, \tau\}\} \quad \text{rpm}. \tag{1}$$

Thus the reference acceleration is 200 rpm/s, or two percentage points of rated speed per second. External PTO load is identically zero. The compressor still consumes shaft power: no external load does not mean zero internal torque.

The plant starts at 10000 rpm, ramps down during absolute time 5–10 s, and settles at 9000 rpm before the evaluation begins at 60 s. Controller handover is at 45 s. All four traces are within 0.1 rpm of 9000 rpm throughout the final five seconds before the test. The requested acceleration occurs at absolute time 60–65 s; simulation continues to 80.01 s. Sea-level, ISA-zero, Mach-zero settings are shared. The reference governor permits 200 rpm/s.

Logged samples are spaced by 0.015 s. The continuous target ramp lasts exactly 5 s; its first post-end sample is at $\tau = 5.01$ s. Maximum logged-reference discrepancy from the equation above is below 2×10^{-12} rpm. Every saved sample passes the existing model-validity flag, and every external-load sample is zero.

3 Controllers held fixed

Controller	K_p	K_i	K_d
Base PI	.025	.050	0
Retuned Poly2 + PID	.025	.140	.00125
Previous PILCO	.06460568	.17102610	0
Smooth PILCO	.02479762	.05385920	0

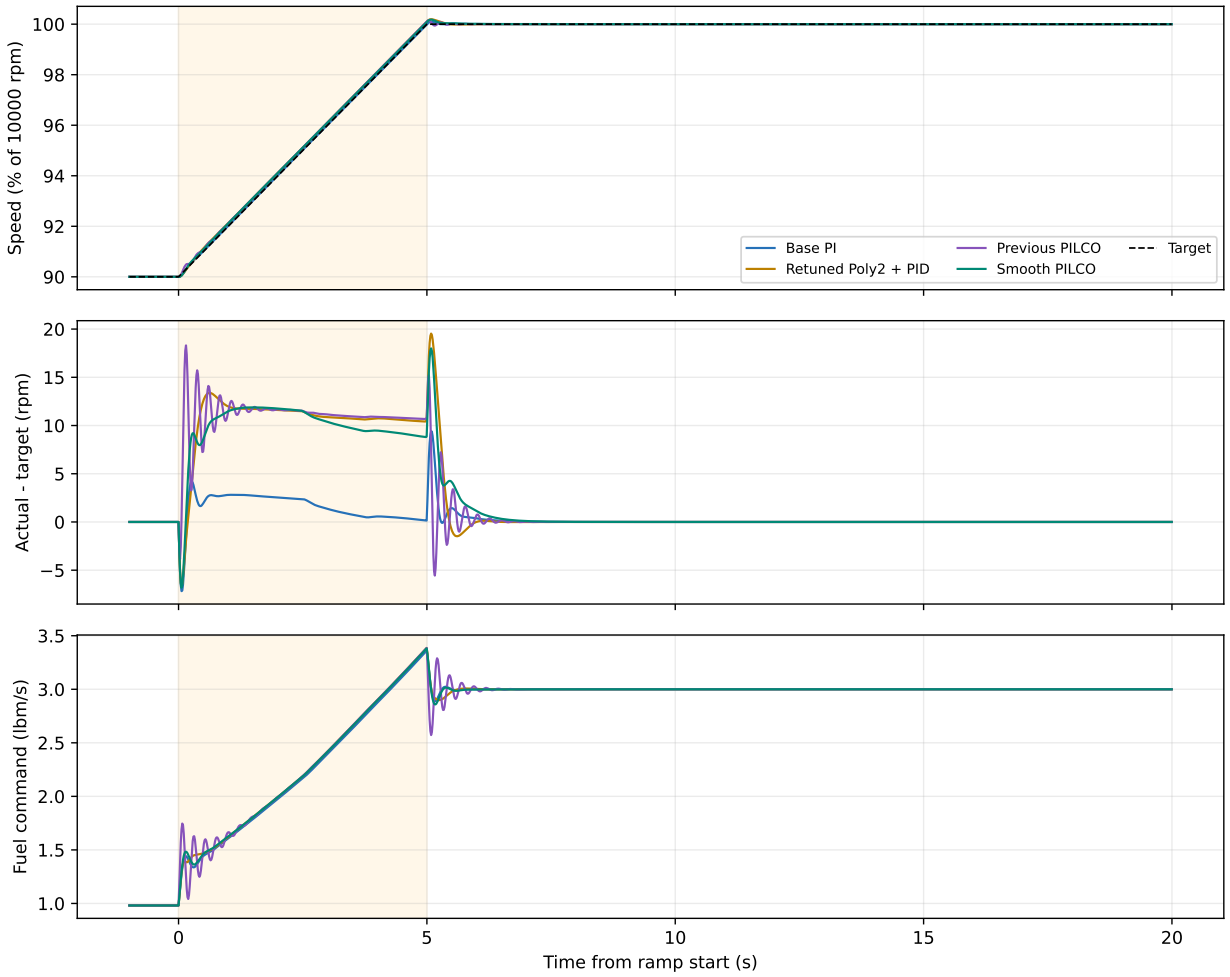
Both PILCO-style policies retain the prior quadratic feedforward, PI feedback, saturation and anti-windup implementation. Their nominal command structure is

$$u_k^* = p_2(r_k, 0, T_k, P_k) + K_p(r_k - N_k) + I_k, \tag{2}$$

$$u_k = \mathrm{clip}(u_k^*, 0.2, 4.0), \qquad I_{k+1} = I_k + hK_i(r_k - N_k), \tag{3}$$

with conditional integration at saturation. The zero second feedforward argument is retained from the prior controller, including during acceleration. No acceleration feedforward term is newly fitted for this test. The Poly2 PID comparator retains its previous inverse-model/derivative implementation; it is not replaced by the PILCO PI law.

4 Measured response



The pale band marks the five-second target ramp. Curves are raw recorded simulation traces; no display smoothing is used. Speed curves nearly overlap at the full-speed scale, so the signed error panel is needed to distinguish tracking performance.

5 Quantitative comparison

For each sample, $e_k = N_k - r_k$. Ramp RMSE uses $0 \le \tau < 5$ s; full RMSE uses $0 \le \tau \le 20.01$ s. Settling is measured after ramp completion: the first recorded time after which every remaining sample stays within ±1 rpm of 10000 rpm, minus 5 s. It is a finite-record measurement.

Controller	Ramp RMSE rpm	Full RMSE rpm	Peak $\lvert e \rvert$ rpm	Settle s after ramp
Base PI	2.161	1.309	9.42	0.610
Retuned Poly2 + PID	10.949	5.799	19.54	0.760
Previous PILCO	11.224	5.710	18.31	0.820
Smooth PILCO	10.252	5.392	18.00	1.045

Fuel excess total variation is

$$R_u = \sum_{k=a+1}^{b} |u_k - u_{k-1}| - |u_b - u_a|. \tag{4}$$

It measures movement beyond net endpoint change, in lbm/s, not fuel consumption. For acceleration, some excess variation is physically useful: fuel may rise to accelerate and then drop to its steady value. Therefore the whole-test measure is not a pure measure of unwanted oscillation. A separate post-ramp value is included. These intervals differ from the earlier load-pulse study; their totals must not be compared directly.

Controller	Whole R_u lbm/s	Post-ramp R_u lbm/s	Overshoot rpm	Min. surge %
Base PI	1.2188	0.2895	9.42	13.98
Retuned Poly2 + PID	1.0004	0.2189	19.54	13.74
Previous PILCO	6.2413	2.6423	15.23	13.74
Smooth PILCO	1.3489	0.3454	18.00	13.78

6 Interpretation and limits

Base PI gives the smallest ramp and full-evaluation RMSE. The smooth PILCO ramp RMSE is 10.252 rpm versus 11.224 rpm for the previous PILCO, an 8.7% reduction. This does not establish a universal ranking: the earlier load-disturbance objective and the present acceleration objective are different.

PILCO gains were learned around 9250 rpm under load pulses. This ramp moves beyond that local operating condition; portions also lie beyond the approximately 9264–9750 rpm polynomial fitting range. The present result is evidence for this single deterministic nominal simulation, not a robustness guarantee. It does not validate the Gaussian-process model across the entire ramp or prove that a ringing penalty alone explains all differences between the gain sets.

The previous derivation remains applicable to how the gains were obtained: expected tracking and fuel-magnitude costs were augmented by expected excess fuel variation, using previous-command and segment-anchor memory. This run does not retrain that objective. A dedicated acceleration-training study would be needed to claim optimized ramp performance.

7 Reproduction

Run `ramp90_equal_sensor_batch.m` in MATLAB with the existing T-MATS dependencies. The four `GasTurbine_EqualSensor_*.mdl` copies preserve previous studies. Raw CSV/MAT traces and summary metrics are in `results/pilco_ramp90_equal_sensor_20261010`. Run `build_ramp90_equal_sensor.py` for this analysis and `build_ramp90_equal_sensor.py` for the synchronized video.